

CricNomics

IPL Player Auction Price Prediction using XGBoost | Team SentiNULLS

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Project Overview

CricNomics predicts the auction value of IPL players using historical performance, international career records, IPL statistics, leadership experience, and player demographics. The XGBoost Regression model captures complex relationships between player attributes and auction prices, helping franchises make informed bidding decisions and optimize team investments. The solution supports data-driven player valuation and strategic auction planning.

Dataset

IPL Player Auction Dataset — 130 players, 26 features:

- Player Age, Country, IPL Team
- Playing Role and Captaincy Experience
- Test Career Statistics
- ODI Career Statistics
- IPL Batting Statistics
- IPL Bowling Statistics
- Batting Average, Strike Rate, Economy Rate
- Base Price and Sold Price

Target Variable: **SOLD PRICE** (Player Auction Price).

Data Preprocessing

- Data cleaning and validation of player records.
- Removal of irrelevant and redundant attributes.
- Encoding of categorical variables such as country, role, and IPL team.
- Feature scaling for numerical attributes where required.
- 80:20 train-test split for unbiased model evaluation.

Model Development

XGBoost Regression was selected because of its high predictive performance, robustness, and ability to model non-linear relationships. The boosting framework sequentially builds decision trees where each new tree corrects errors made by previous trees. Hyperparameters such as learning rate, maximum depth, and number of estimators were optimized to improve generalization and prediction accuracy.

Model Evaluation

The model was evaluated using:

- R-Squared Score
- Mean Absolute Error (MAE)
- Mean Squared Error (MSE)
- Root Mean Squared Error (RMSE)
- Feature Importance Analysis

Results

- Playing Role emerged as a major factor influencing player valuation.
- Batting Average and Strike Rate strongly affected auction prices.
- International career statistics increased player demand and market value.
- Captaincy experience positively influenced premium player valuations.
- Feature importance analysis highlighted performance consistency as a key pricing factor.

Business Impact

- Estimate player value before auction events.
- Optimize franchise budget allocation.
- Identify undervalued players with strong potential.
- Support data-driven bidding strategies.
- Improve squad composition and auction planning.
- Provide a competitive advantage through predictive analytics.

Conclusion

CricNomics demonstrates the successful application of machine learning in sports analytics. Using XGBoost Regression, the system effectively predicts IPL player auction prices from performance and career-based attributes. The project showcases how predictive analytics can improve decision-making, maximize investment efficiency, and enhance team-building strategies in professional cricket auctions.

Project Summary

Team Name: SentiNULLS

Project Name: CricNomics

Model Used: XGBoost Regression

Dataset: IPL Player Auction Dataset (130 × 26)

Target Variable: SOLD PRICE

Team Members: Aryan Shukla, Chetan Khera, Kanishk Singh, Mihir Dixit, Shashank Singh